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ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-1** **Date: 26-03-2026**

Time: 09.00AM to 12.00PM **GTM-29** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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26-03-2026_Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main-Shift-1_GTM-29_Test Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let the solution set of the equation $\sqrt{[x]+\left[\frac{x}{2}\right]+\left[\sqrt{\{x\}}+\left[\frac{x}{3}\right]\right]}=3$ is [a, b) then product ab = (where [.] and { } denote greatest integer function and fractional part function)

1) 9 **2)** 12 **3)** 15 **4)** 18

2. The value of $\underset{n\to\infty}{Lt}\left(\frac{n!}{n^n}\right)^{\frac{3n^3+4}{4n^4-1}}, n\in N$ is equal to

1) $e^{-3/4}$ **2)** $e^{-4/3}$ **3)** e^{-4} **4)** $e^{-1/4}$

3. The least integer which is greater than or equal to the area of region in xy plane satisfying $x^6-x^2+y^2\le 0$ is

1) 1 **2)** 2 **3)**3 **4)** 4

4. The remainder when $2^{30}\times 3^{20}$ is divided by 7 is

1) 1 **2)** 6 **3)** 4 **4)** 2

5. Consider a square matrix A of order 2 which has its elements as 0, 1, 2 and 4. Let N denote the number of such matrices

	COLUMN-I		COLUMN-II
A)	Possible non –negative values of det (A) is	**P)**	2
B)	Sum of values of determinants corresponding to N matrices is	**Q)**	4
C)	If absolute value of det (A) is least, then possible value of |adj (adj(adj A))|	**R)**	-2
D)	If det (A) is least, then possible value of det $\left(4A^{-1}\right)$ is	**S)**	0
		T)	8

1) A – P,Q,T, B- T, C-P, D-P **2)** A – P,Q, B- S, C-R, D-R

3) A – P,Q,T, B- S, C-P,R, D-R **4)** A – Q,T, B- T, C-P,R, D-P

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6. Statement 1: The sum of all the 4 –digited distinct numbers that can be formed with digits 1,3,3,4 is 36663

Statement 2: The number of rectangles in a chess board which are not squares is 1092

1) Statement 1 is TRUE and Statement 2 is TRUE

2) Statement 1 is FALSE and Statement 2 is FALSE

3) Statement 1 is TRUE and Statement 2 is FALSE

4) Statement 1 is FALSE and Statement 2 is TRUE

7. If $2^x = 3^y = 6^{-z}$ then value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=$

1) -3 **2)** 2 **3)** 1 **4)** 0

8. The minimum value of $y=(\sin x+\cos ecx)^2+(\cos x+\sec x)^2+(\tan x+\cot x)^2$ where $x \in R$ is

1) 9 **2)** 15 **3)** 13 **4)** 7

9. The number of real values of x satisfying the equation $3\sin^{-1}x+\pi x-\pi=0$ is /are

1) 0 **2)** 1 **3)** 2 **4)** 3

10. 3 persons independently solve a certain problem correctly with probabilities $\frac{1}{2},\frac{1}{3},\frac{1}{4}$ respectively. Then the probability that the problem is solved correctly by at least one of them is

1) $\frac{1}{24}$ **2)** $\frac{1}{26}$ **3)** $\frac{1}{2}$ **4)** $\frac{3}{4}$

11. Number of positive integral solution of $x_1x_2x_3=120$

1) 27 **2)** 90 **3)** 60 **4)** 150

12. Consider the function $f(x)=\begin{cases}\frac{P(x)}{\sin(x-2)} & , \quad x\neq 2\\ 7 & , \quad x=2\end{cases}$ where P(x) is a polynomial such that $P^{11}(x)$ is always a constant and P(3) = 9. If f(x) is continuous at x = 2 then P(4) =

1) 11 **2)** 10 **3)** 22 **4)** 20

13. The shortest distance between the line x – y = 1 and the curve $x^2=2y$ lies in the interval

1) $(0.1,0.2)$ **2)** $(0.2,0.3)$ **3)** $(0.3,0.4)$ **4)** $(0.4,0.5)$

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14. Consider two vectors $\bar{u} = 3\bar{i} - \bar{j}$ *and* $\bar{v} = 2\bar{i} + \bar{j} - \lambda\bar{k}, \lambda > 0$. Angle between them is $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$. Let $\bar{v} = \bar{v}_1 + \bar{v}_2$, where $\bar{v}_1$ is parallel to $\bar{u}$ and $\bar{v}_2$ is perpendicular to $\bar{u}$. Let value of $|\bar{v}_1|^2 + |\bar{v}_2|^2$ be k. Where $k \in N$. Then number of positive divisors of k are

1) 4 **2)** 14 **3)** 7 **4)** 2

15. Let the values of λ, for which the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-\lambda}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{6}}$ be λ_1 *and* λ_2. Then Area of circle passing through points $(0,0),(\lambda_1,\lambda_2)$ *and* (λ_2,λ_1) is

1) 16π **2)** $\frac{50\pi}{9}$ **3)** 9π **4)** $\frac{2\pi}{9}$

16. Let $f:[1,\alpha) \to R$ be a differentiable function such that $f(1) = \frac{1}{3}$ and $3\int_1^x f(t)\,dt = x f(x) - \frac{x^3}{3}, x \in [1,\infty)$ then f (e) =

1) $\frac{e^2+4}{2}$ **2)** $\frac{\log_e 4 + e}{3}$ **3)** $\frac{4e^2}{3}$ **4)** $\frac{e^2-4}{3}$

17. The value of $\int_0^{1/2} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{1/4}}$ dx is

1)2 **2)**4 **3)**3 **4)** 1

18. Circles of radii 36 and 9 touches externally. The radius of the circle which touches the two circles externally and also their direct common tangent is

1) $3\sqrt{5}$ **2)**18 **3)**4 **4)** 2

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19. If the function $y = e^{4x} + 2e^{-x}$ is a solution of the differential equation

$$\frac{\frac{d^3y}{dx^3} - 13\frac{dy}{dx}}{2y} = k, k \in N$$ then, number of solutions of θ such that $\tan\theta$ =k

when $\theta \in (-\pi, \pi)$ is

1) 1 **2)** 0 **3)** 4 **4)** 2

20. If the sum of all the roots of the equation $e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ is logP then P =

1) 45 **2)** 81 **3)** 9 **4)** 18

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Suppose AB is a focal chord of the parabola $y^2 = 12x$ of length ℓ and slope $m < \sqrt{3}$.

If the distance of chord AB from origin is d then ℓd^2 =

22. Let z be a complex number with non –zero imaginary part. If $\frac{2 + 3z + 4z^2}{2 - 3z + 4z^2}$ is a real

number then value of $20|z|^2$ is

23. If A is a 2x2 matrix such that trace of A is 3 and trace of A^3 is -18, then the value of determinant of A is

24. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in G.P and the

arithmetic mean of a, b, c is b + 2, then value of a+1 is

25. If xdy = y(dx + ydy) , y(1) =1 and y(x) > 0 then value of y(-3) is

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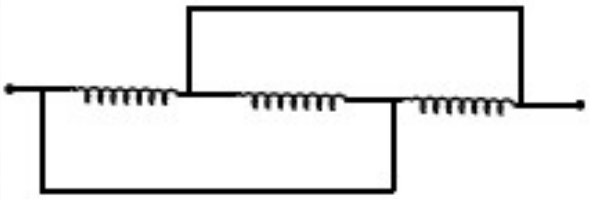
PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

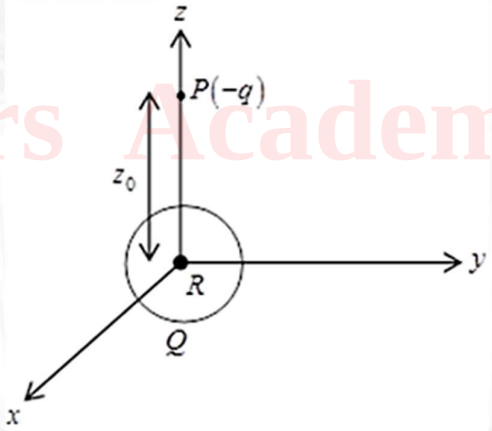
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. Pure inductance of each 3.0H is connected as shown below. The equivalent inductance of the circuit is



1) 1H **2)** 2H **3)** 3 H **4)** 9H

27. A positively charged thin metal ring of radius R is fixed in the $x-y$ plane with its centre at the origin O. A negatively charged particle P is released from rest at the point $(0,0,z_0)$ where $z_0 > 0$. Then the motion of P is :



1) Periodic for all value of z_0 satisfying $0 > z_0 < \infty$

2) Simple harmonic for all value of z_0 satisfy $0 < z_0 \le R$

3) Approximately simple harmonic provided $z_0 << R$

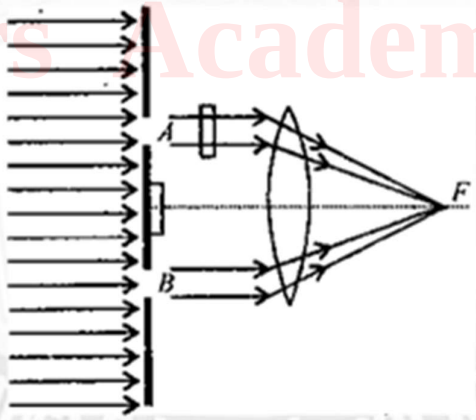
4) Such that P crosses O and continues to move along the negative z axis towards $z = -\infty$

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28. A steam boat goes across a lake and comes back (i) on a quiet day when the water is still and (ii) on a rough day when there is a uniform current so as to help the journey onward and to impede the journey back. If the speed of the launch, on both days, was same, the time required for the complete journey on the rough day as compared to that on the quiet day, will be

1) less **2)** Same **3)** more **4)** cannot be predicted

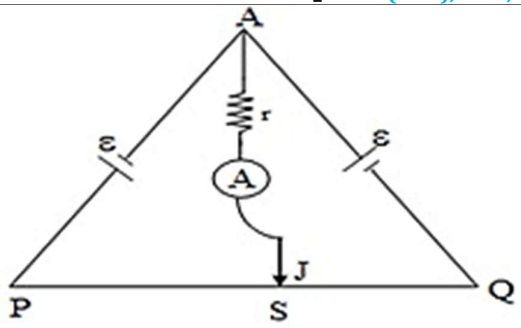
29. In a modified Young's double slit experiment, a monochromatic uniform and parallel beam of light of wavelength $6000\overset{0}{A}$ and intensity $(10/\pi)Wm^2$ is incident normally on two circular apertures A and B of radii 0.001m and 0.002m respectively. A perfectly transparent film of thickness $2000\overset{0}{A}$ and refractive index 1.5 for the given wavelength is placed in front of aperture A, see fig. Calculate the power (in watt) received at the focal spot F of the lens. The lens is symmetrically placed with respect to the apertures. Assume that 10% of the power received by each aperture goes in the original direction and is brought to the focal spot.



1) $7\times10^{-6}W$ **2)** $7\times10^{-5}W$ **3)** $8\times10^{-6}W$ **4)** $7\times10^{-8}W$

30. In the figure, PQ is a wire of uniform cross-section and resistance R_0. A is an ideal ammeter and the cells are of negligible internal resistance. The jockey J can be freely slide over wire PQ making contact on it at S. Length of wire PQ is l.

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Reading of ammeter when $PS = \frac{l}{n}$ is

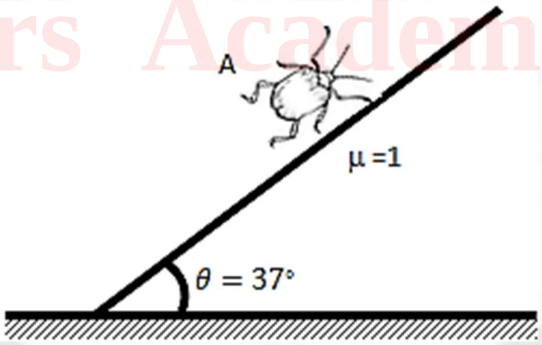
1) $\left(1-\frac{4}{n}\right)\frac{\varepsilon}{\left(r+\frac{R_0}{n}\right)}$

2) $\left(1-\frac{2}{n}\right)\frac{\varepsilon}{\left(r+\frac{R_0}{n}\right)}$

3) $\left(1-\frac{1}{n}\right)\frac{\varepsilon}{\left(r+\frac{R_0}{n}\right)}$

4) $\left(1-\frac{1}{n^2}\right)\frac{\varepsilon}{\left(r+\frac{R_0}{n}\right)}$

31. An insect of mass m, starts moving on a rough inclined surface from point A. As the surface is very sticky, the coefficient of friction between the insect and the incline is $\mu = 1$. Assume that it can move in any direction, up the incline or down the incline then



1) The maximum possible acceleration of the insect can be $14\,m/s^2$ down the incline plane

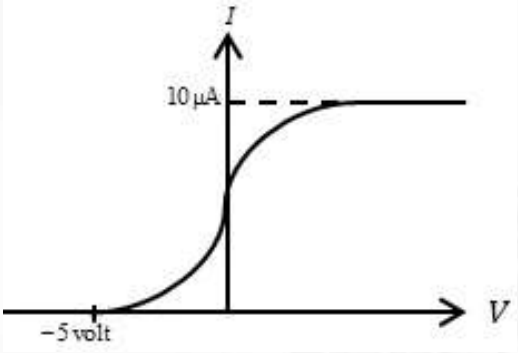
2) The maximum possible acceleration of the insect can be $4m/s^2$ up the incline plane

3) The insect can move with velocity $5m/s^2$

4) The insect cannot move with a constant velocity

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32. In the photoelectric experiment, if we use a monochromatic light, the $I-V$ curve is as shown. If work function of the metal is 2 eV, estimate the power of light used (in W). Assume efficiency of photo emission to be $10^{-3}\%$, i.e., number of photoelectrons emitted are $10^{-3}\%$ of number of photons incident on metal.

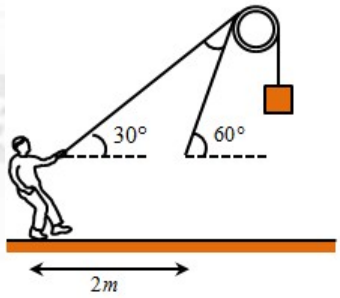


1) 1 W **2)** 2 W **3)** 4 W **4)** 7 W

33. A solid body rotates about a stationary axis so that its angular velocity depends on the rotational angle ϕ as $\omega=\omega_0-k\phi$ where ω_0 and k are positive constants. At the moment $t=0, \phi=0$, the time dependence of rotation angle is

1) $k\omega_0 e^{-kt}$ **2)** $\frac{\omega_0}{k}e^{-kt}$ **3)** $\frac{\omega_0}{k}\left(1-e^{-kt}\right)$ **4)** $\frac{k}{\omega_0}\left(e^{-kt}-1\right)$

34. A person lifts a 25kg block hanging over a fixed light frictionless small pulley by walking horizontally, as shown in figure. As the person walks 2m, the angle of the rope to the horizontal changes from 60° *to* 30°. If the block rises at constant speed, the work done by rope on the person as the person moves by $2m$ is (consider the rope to be light and inextensible) (take $g=10\,\text{m/s}^2$)



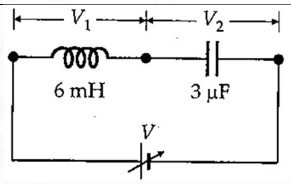
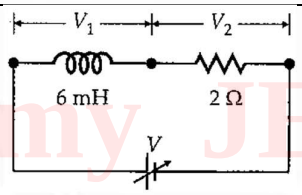
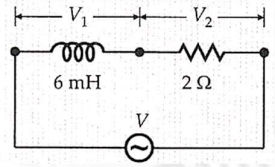
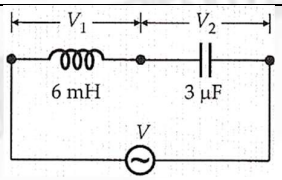
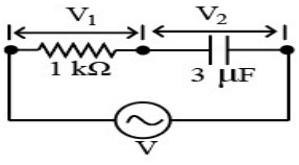
1) $500\left(\sqrt{3}-1\right)$J **2)** $500\left(1-\sqrt{3}\right)$ J **3)** $500\sqrt{3}$ J **4)** None of these

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35. A stone is hung in air from a wire which is stretched over a sonometer. The bridges of the sonometer are 40 cm apart when the wire is in unison with a tuning fork of frequency 256 Hz. When the stone is completely immersed in water, the length between the bridges is 22 cm for reestablishment of unison. The specific gravity of the material of the stone is

1) $\frac{(40)^2}{(40)^2+(22)^2}$ **2)** $\frac{(40)^2}{(40)^2-(22)^2}$ **3)** $256\times\frac{22}{40}$ **4)** $256\times\frac{40}{22}$

36. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in Column II. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltages V1, and V2 (indicated in circuits) are related as shown in Column I.
Match the two.

	COLUMN-I		COLUMN-II
A)	$I \neq 0$, V_1 is proportional to I	**P)**	
B)	$I \neq 0, V_2 > V_1$	**Q)**	
C)	$V_1 = 0, V_2 = V$	**R)**	
D)	$I \neq 0$, V_2 is proportional to I	**S)**	
		T)	

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1) A – P, Q, R, T ;B- S,T ; C-P, R ;D- P, R, S

2) A – P,Q, S; B- P, R, S,T ;C- P,Q ; D- Q, S,T

3) A – Q, R, S,T ; B- P,R ; C- P,R,S ; D- Q, R,T

4) A – R,S,T; B- Q,R,S,T;C- P,Q ;D- Q,R,S,T

37. During the propagation of electromagnetic waves in a medium

1) Electric energy density is double of the magnetic energy density

2) Electric energy density is half of the magnetic energy density

3) Electric energy density is equal to the magnetic energy density

4) Both electric and magnetic energy densities are zero

38. Which is incorrect statement regarding reverse saturation current in p-n junction diode?

1) Current is due to minority carriers

2) Current doubles for every 100 °C rise in temperature

3) Current carriers are produced by thermal agitation

4) Reverse saturation current is also known as leakage current

39. Assertion. Soft iron is used as a core of transformer.

Reason. Area of hysteresis loop for soft iron is small.

1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)

2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)

3) (A) is true but (R) is false

4) Both (A) and (R) are false

40. Assertion: As temperature of a black body is raised, wavelength corresponding to maximum energy reduces.

Reason: Higher temperature would mean higher energy and hence higher wavelength.

1) If both assertion and reason are true and reason is the correct explanation of the assertion.

2) If both assertion and reason are true but reason is not correct explanation of the assertion.

3) Assertion is true, but reason is false.

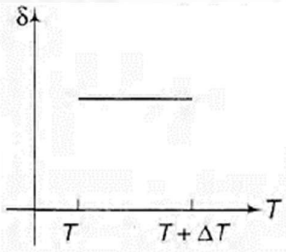
4) Assertion is false and reason is true.

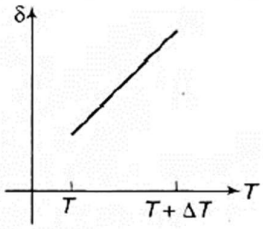
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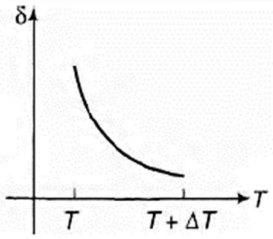
41. Statement-1. When ultraviolet light is incident on a photocell, its stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is K_{max}. When the ultraviolet light is replaced by X-rays, both V_0 and K_{max} increase.
Statement-2. Photoelectrons are emitted with speeds ranging from zero to a maximum value because of the range of frequencies present in the incident light.
1) Statement-1 is true, Statement-2 is false
2) Statement-1is true,statement-2 is true; statement-2 is the correct explanation of statement-1
3) Statement-1 is true, statement-2 is true; statement-2 is the not the correct explanation of statement-1
4) Statement-1 is false, Statement-2 is true

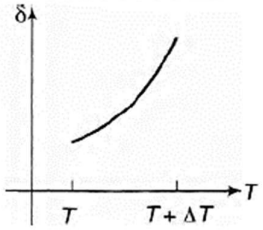
42. Statement-1: A metallic surface is irradiated by a monochromatic light of frequency $v > v_0$ (the threshold frequency). The maximum kinetic energy and the stopping potential are K_{max} and V_0 respectively. If the frequency incident on the surface is doubled, both the Kmax and V_0 are also doubled.
Statement-2: The maximum kinetic energy and the stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light.
1) Statement-1 is true, Statement-2 is false
2) Statement-1is true,statement-2 is true; statement-2 is the correct explanation of statement-1
3) Statement-1 is true, statement-2 is true; statement-2 is the not the correct explanation of statement-1
4) Statement-1 is false, Statement-2 is true

43. An ideal gas is initially at temperature T and volume V. Its volume is increased by ΔV due to an increase in temperature ΔT, pressure remaining constant. The quantity $\delta = \Delta V / V\Delta T$ varies with temperature as

1)


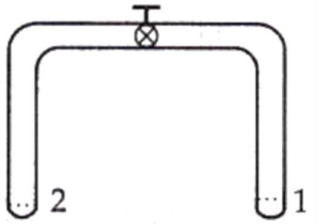
2)


3)


4)


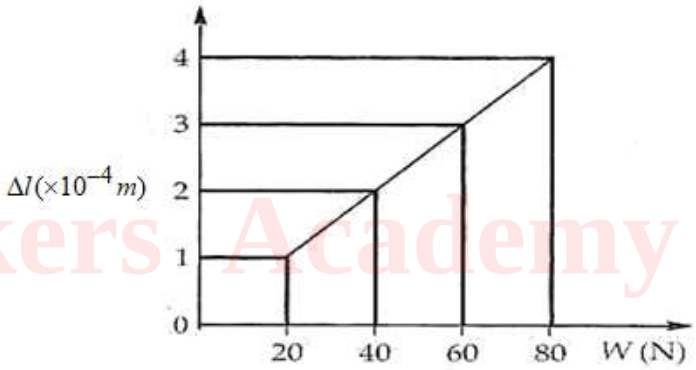
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44. A glass tube of uniform internal radius (r) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius r. End 2 has sub-hemispherical soap bubble as shown in figure. Just after opening the valve,



1) Air from end 1 flows towards end 2. No change in the volume of the soap bubbles.
2) Air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases.
3) No changes occur.
4) Air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases.

45. The following graph shows the extension (Δl) of a wire of length 1 m suspended from the top of a roof at one end and with a load W connected to the other end.



If the cross-sectional area of the wire is $10^{-6} m^2$, calculate the Young's modulus of the material of the wire

1) 2×10^{11} N / m^2 **2)** 2×10^{-11} N / m^2 **3)** 3×10^{-12}N/m^2 **4)** 2×10^{-13}N/m^2

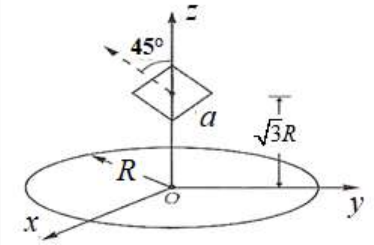
SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

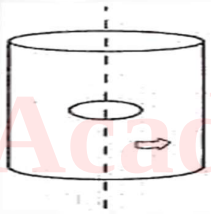
46. A bomb is fired from a canon with a velocity of 1000 m/s making an angle of 30° with the horizontal $\left(g = 9.8 m / s^2\right)$.Time taken by bomb to reach the highest point is------s

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47. A circular wire loop of radius R is placed in the x-y plane centered at the origin O. A square loop of side a (a << R) having two turns is placed with its centre at a distance $\sqrt{3}R$ along the axis of the circular wire loop, as shown in the figure. The plane of the square loopmakes an angle of 45° with respect to the z-axis. If the mutual inductance between the loops is given by $\frac{\mu_0 a^2}{2^{p/2} R'}$ then the value of p is _____ .



48. A long circular tube of length 10 m and radius 0.3 m carries a current I along its curved surface as shown. A wire-loop of resistance 0.005 ohm and of radius 0.1 m is placed inside the tube with its axis coinciding with the axis of the tube. The current varies as $I = I_0 \cos(300\,t)$, where I_0 is constant. If the magnetic moment of the loop is $N\mu_0 I_0 \sin(300\,t)$, then find N.



Rankers Academy JEE

49. A proton is fired from very far away towards a nucleus with charge Q = 120e, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is_____.

[take $m_p = (5/3)\times10^{-27}\,kg; h/e = 4.2\times10^{-15}\,Js/C; \frac{1}{4\pi\varepsilon_0} = 9\times10^9\,m/F; 1\,fm = 10^{-15}\,m$

50. During Searle's experiment, zero of the Vernier scale lies between 3.20×10^{-2}m and 3.25×10^{-2}m of the main scale. The 20th division of the Vernier scale exactly coincides with one of the main scale divisions. When an additional load of 2 kg is applied to the wire, the zero of the Vernier scale still lies between 3.20×10^{-2} m and 3.25×10^{-2} of the main scale but now the 45th division of Vernier scale coincides with one of the main scale divisions. The length of the thin metallic wire is 2 m and its cross-sectional area is 8×10^{-7}m^2. The least count of the Vernier scale is 1.0×10^{-5}m. The maximum percentage error in the Young's modulus of the wire is _____

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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. A sample of $CaCO_3$ and $MgCO_3$ weighed 2.21g is ignited to constant weight of 1.152g. The composition of mixture is: (Given molar mass in g mol^{-1} $CaCO_3$:100. $MgCO_3$:84)

1) $1.187gCaCO_3 + 1.023gMgCO_3$ **2)** $1.023gCaCO_3 + 1.023gMgCO_3$

3) $1.187gCaCO_3 + 1.187gMgCO_3$ **4)** $1.023gCaCO_3 + 1.187gMgCO_3$

52. The bond dissociation energy of B –F in BF_3 is 646 kJ mol^{-1}, whereas that of C-F in CF_4 is 515 kJ mol^{-1}. The correct reason for higher B –F bond dissociation energy as compared to that of C-F bond is

1) Stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4.

2) Significant $p\pi - p\pi$ interaction between B and F in BF_3, whereas there is no possibility of such interaction between C and F in CF_4.

3) Lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4.

4) Smaller size of B –atom as compared to that of C –atom.

53. Arrange the elements with the following elements with the following electronic configurations in increasing order of electron affinity

(i) $1s^2s^22p^5$ (ii) $1s^22s^22p^4$ (iii) $1s^22s^22p^63s^23p^4$ (iv) $1s^22s^22p^63s^23p^5$

1) $(ii) < (iii) < (i) < (iv)$ **2)** $(iii) < (ii) < (iv) < (i)$

3) $(iii) < (ii) < (i) < (iv)$ **4)** $(ii) < (iii) < (iv) < (i)$

54. HI is a covalent compound in which small positive charge is present on H and a small negative charge on I, and the molecule is represented as $H^{\delta+} - I^{\delta-}$ the dipole moment of HI is 0.38D and the bond length. H -I is 1.61A°. What fraction of charge is present on I ?

1) 10% **2)** 20% **3)** 5% **4)** 80%

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55. Statement -1 : $\left[Fe(CN)_6\right]^{-4} \rightarrow Fe^{3+} + CO_2 + NO_3^-$, the equivalent weight of reactant is 3.47.

Statement -2 : Equivalent weight of reactant $= \frac{Molecular\ weight}{61}$

1) Statement-1 is true, Statement-2 is false

2) Statement-1is true, statement-2 is true; statement-2 is the correct explanation of statement-1

3) Statement-1 is true, statement-2 is true; statement-2 is the not the correct explanation of statement-1

4) Statement-1 is false, Statement-2 is true

56.

	COLUMN-I		COLUMN-II
A)	H_2S	**P)**	Decolourises acidified sol. of $KMnO_4$
B)	SO_2	**Q)**	Disproportionation reaction
C)	NO_2	**R)**	Bleaching action
D)	HNO_2	**S)**	V – shaped structure

1) A – P,S, B- P,Q,R,S C-P,S D-P,S **2)** A – P,Q, B- S, C-R, D-R

3) A – P,Q,T, B- S, C-P,R, D-R **4)** A – Q,T, B- T, C-P,R, D-P

57. The correct reason (s) for the acidic nature of an aqueous solution of aluminium chloride is that

1) For aluminium ions the, $\frac{charge}{surface\ area}$, ratio is large

2) The O–H bonds in $\left[Al(H_2O)_6\right]^{3+}$ are is weaker than O-H bond in water

3) Aluminium chloride is covalent compound and its aqueous solution is acidic.

4) Both 1 & 2

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58. Bromine (molecular mass.160 amu) boils at 59°C, whereas iodine monochloride (molecular mass. 162 amu) boils at 92°. The best explanation for this difference in b.p. is

1) The *I-Cl* bond is stronger than Br-Br bond

2) Br_2 is a non –polar molecule, *ICl* is polar

3) The vander waal's forces in *ICl* are stronger than in Br_2

4) The difference in molecular masses

59. Which of the following will have two stereoisomeric forms?

(I) $\left[Cr(NO_3)_3(NH_3)_3\right]$ (II) $K_3\left[Fe(C_2O_4)_3\right]$

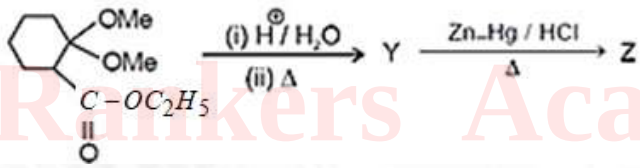
(III) $\left[NiCl_2(en)_2\right]^+$ (IV) $\left[NiBrCl(Ox)_2\right]^{3-}$

1) I only **2)** I and II **3)** III and IV **4)** All of these

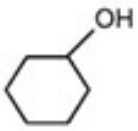
60. $CHD_2I \xrightarrow{Mg} \xrightarrow{CH_2=O} \xrightarrow{D^\oplus/D_2O} \xrightarrow{Al_2O_3.\Delta} E$. The product E' is ':

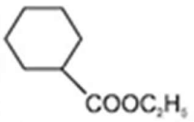
1) $CHD=CH_2$ **2)** $CH_2=CH_2$ **3)** $CD_2=CD_2$ **4)** $CH_2=CD_2$

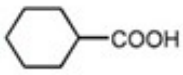
61.



Finally obtained product is

1) 

2) 

3) 

4) 

62. Which of the following reactions of glucose can be explained only by its cyclic structure?

1) Glucose forms penataacetate.

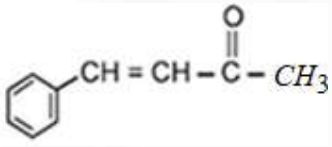
2) Glucose reacts with hydroxylamine to form an oxime.

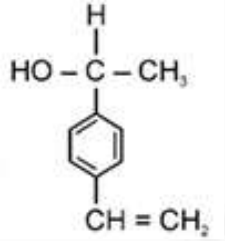
3) Pentaacetate of glucose does not react with hydroxylamine.

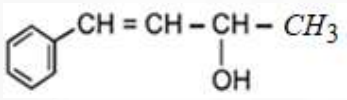
4) Glucose is oxidized by nitric acid to gluconic acid

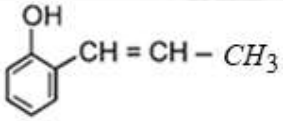
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63. Compound '$P'(C_{10}H_{12}O)$ evolves H_2 with Na metal. If reacts with Br_2 / CCl_4 to give 'Q' $(C_{10}H_{12}Br_2O)$. P on reaction with $I_2 / NaOH$ it forms iodoform and an acid 'R' $(C_9H_8O_2)$.'P' has geometrical and optical isomers. The structure of 'P' should be :

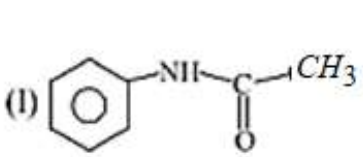
1) 

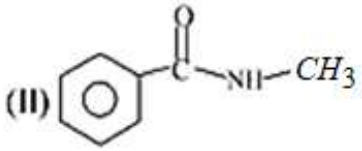
2) 

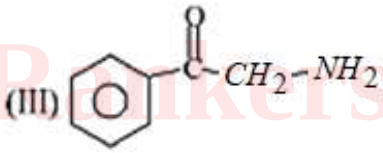
3) 

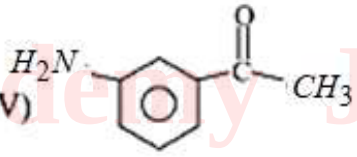
4) 

64. The correct basic strength order of the following bases is:









1) $I > II > IV > III$ **2)** $IV > III > II > I$ **3)** $III > II > IV > I$ **4)** $III > IV > II > I$

65. Identify the group that contains colourless ions.

1) La^{3+} *and* Lu^{3+} **2)** Eu^{3+} *and* Tb^{3+}

3) Ce^{3+} *and* Lb^{3+} **4)** Tm^{3+} *and* Yb^{3+}

66. Identify the compound, which gives negative test with ninhydrin and positive test with Benedict's solution?

1) An amino acid **2)** A monosaccharide **3)** A protein **4)** A lipid

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67. Identify correct statement/s :

(A) $-OCH_3$ *and* $-NHCOCH_3$ are activating group.

(B) $-CN$ *and* $-OH$ are meta directing group.

(C) – CN and SO_3H are meta directing group.

(D) Activating groups act as ortho – and para directing groups.

(E) Halides are activating groups.

Choose the correct answer from the options given below:

Options:

1) (A), (C) and (D) only **2)** (A), (B) and (E) only

3) (A) and (C) only **4)** (A) only

68. Assertion (A): Aryl iodides can be prepared by reaction of arenes with iodine in the presence of an oxidizing agent.

Reason (R): Oxidising agent oxidizes I_2 into HI.

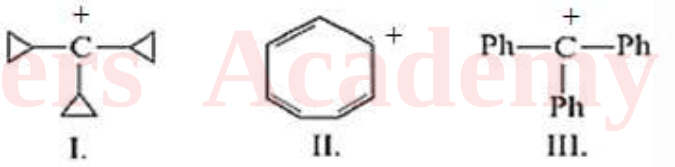
1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)

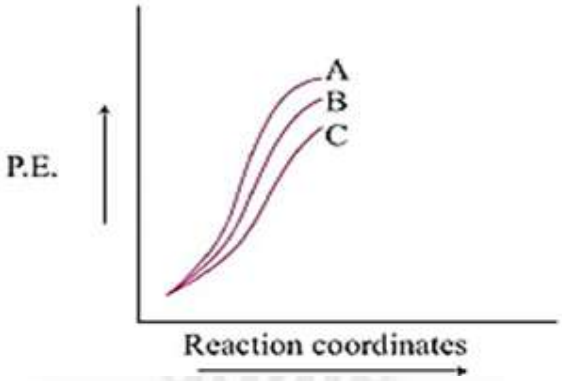
2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)

3) (A) is true but (R) is false

4) Both (A) and (R) are false

69. In the following graph, stability of different carbocations have been shown:





Match the potential energy curve with carbocation.

1) $\begin{matrix} I & II & III \\ A & B & C \end{matrix}$ **2)** $\begin{matrix} I & II & III \\ B & A & C \end{matrix}$ **3)** $\begin{matrix} I & II & III \\ C & B & A \end{matrix}$ **4)** $\begin{matrix} I & II & III \\ C & A & B \end{matrix}$

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70. Match column -I with Column-II

	COLUMN-I		COLUMN-II
A)	Ammonolysis	**p)**	Amine with lesser number of carbon atoms than reactant
B)	Gabriel phthalimide synthesis	**q)**	Detection test for primary amines.
C)	Hoffmann bromamide reaction	**r)**	Reaction of phthalimide with KOH and R -X
D)	Carbylamine reaction	**s)**	Reaction of alkythalids with NH_3

1) $A-s, B-r, C-P, D-q$ **2)** $A-r, B-p, C-r, D-s$

3) $A-q, B-r, C-s, D-p$ **4)** $A-s, B-p, C-q, D-r$

SECTION-II
(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71 What would be the electrode potential for the given half cell reaction at pH = 5?

$2H_2O \rightarrow O_2 + 4H^+ + 4e^-; E^0_{red} = 1.23V$

($R = 8.314 J\,mol^{-1}K^{-1}; Temp = 298K$; oxygen under std. atm. Pressure of 1 bar)

72. When $Fe_{0.93}O$ is heated in presence of oxygen, it converts to Fe_2O_3. The number of Correct Statement/s from the following is _____

1) The equivalent weight of $Fe_{0.93}O$ is $\frac{Molecular\,Weight}{0.79}$

2) The number of moles of Fe^{2+} and Fe^{3+} in 1 mole of $Fe_{0.93}O$ is 0.79 and 0.14 respectively.

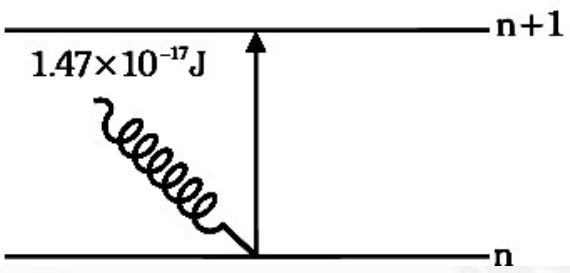
3) $Fe_{0.93}O$ is an example of non stoichiometric compound

4) The % composition of Fe^{2+} and Fe^{3+} in $Fe_{0.93}O$ is 85% and 15% respectively.

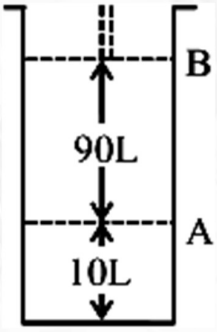
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73. The electron in the nth orbit of Li^{2+} is excited to (n+1) orbit using the radiation of energy 1.47×10^{-17} J (as shown in the diagram). The value of n is _____

Given $R_H = 2.18 \times 10^{-18}$ J



74. Consider the figure provided.

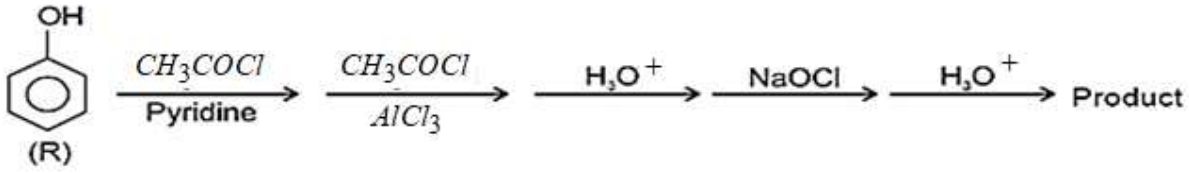


1 mol of an ideal gas is kept In a cylinder, fitted with a piston, at the position A, at 18^0 C. If the pistion is moved to position B, Keeping the temperature unchanged, then 'x' L atm. Magnitude of work done in this reversible process.

X =_____L atm.(nearest integer)

[Given : Absolute temperature = $^0C + 273.15$,R=0.08206L atm $mol^{-1}K^{-1}$]

75.



Molecular mass of major product of this reaction is:

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